Autonomous mobile robot for fruit harvesting using machine vision

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***Abstract* — This paper presents a robotic system based on a tracked mobile unit, equipped with a three-degree-of-freedom manipulator and distance sensors, designed for the autonomous harvesting of agricultural products in simulated rural environments. Perception is achieved using a YOLOv8n network trained on a reduced database to detect a tree, apples, and a harvesting basket. The mobile unit's navigation relies on a differential kinematic controller with potential fields attracting the targets and repulsive fields derived from distance sensors, enabling safe movement between obstacles. The manipulator is controlled by a visual feedback loop that directly adjusts the joints based on the fruit projection error in the image. In simulation tests, the system travels an average of 14.7 m in 7.3 min, transferred all reachable apples (3 of 5 detected, 2 were out of reach), with an overtime of approximately 14% associated with conservative speed settings for operational safety. The inference times of 25 ms and the rate of 34 FPS confirm the feasibility of real-time operation and provides a basis for future real-world validation.**

Keywords —Potential fields, autonomous navigation, YOLO, webots, agricultural mobile robotics.

I. INTRODUCTION

In recent years, multiple robotic platforms have been developed for precision agriculture [1][2], focusing on tasks such as crop monitoring, targeted spraying, and selective harvesting using mobile ground robots with handling tools [3][4]. These systems often integrate advanced perception techniques, including artificial intelligence algorithms such as YOLO, Faster RCNN, or SSD object detection and localization networks, capable of identifying individual fruits and environmental elements under varying lighting and occlusion conditions [5][6]. Simultaneously, widely used navigation approaches in mobile robotics include potential field-based controls [7][8], trajectory planning based on occupancy maps, and obstacle avoidance schemes supported by RGB-LiDAR distance sensors or RGB-D depth cameras [9][10][11]. Furthermore, in agricultural handling, robotic arms are employed that combine classic kinematic control with computer vision strategies to obtain a clear view of the fruit and execute robust approach and grasping movements [12][13]. However, many studies focus on isolated components such as perception, navigation, or manipulation, and struggle to integrate all subsystems in a modular fashion into a unified architecture that can function autonomously in unfamiliar environments [14][15][16]. The main contribution of this work is not a new individual algorithm, but a modular, fully integrated perception, navigation and manipulation architecture for tracked mobile manipulators, validated in simulation. The architecture is designed to be ported to an embedded platform.

II. METHODOLOGY

The methodology presented below corresponds to the development and evaluation of a mobile robotic system using speed and position controls, making use of visual identification algorithms in a simulation environment such as Webots, addressing the problem of identification, acquisition and relocation of agricultural products, with application in rural production areas. To achieve this, four stages in methodological development are defined below.

The first stage involves selecting perception and detection models applicable to the environment, allowing the identification of three specific elements within the workspace: the fruit tree, the fruit (in this case, apples), and the basket where the fruit is placed. The second stage consists of developing the mobile robotic system, along with navigation and obstacle avoidance algorithms, enabling the robot to move continuously and safely to the designated objects. The third stage focuses on designing a visual feedback controller for the manipulator to ensure alignment at the joints, centering the target with the gripping tool. This stage also addresses the transformations performed to obtain global coordinates for the detected objects. Finally, the last stage involves developing a finite state machine (FSM) that integrates all previous stages into the complete system's operation, offering modular functionality and scalability to similar applications.

III. MODEL FOR THE PERCEPTION AND DETECTION OF OBJECTIVES

For the robot's visual perception, a YOLO network is used, specifically the yolov8n version. This is a convolutional network that locates and classifies objects in images, enclosing these detections in boxes called bounding boxes. This network was trained to distinguish between three objects (an apple, tree and a basket), using a simulated environment database of approximately 80 images, distributed in a 70/30 ratio for subsequent training. Fig. 1 shows a sample of this database.

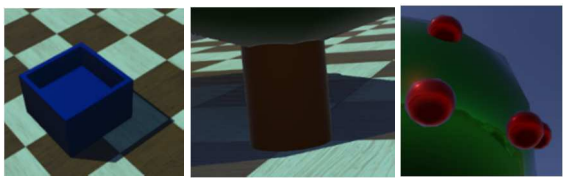


Fig. 1. Sample of simulated image dataset used to train YOLOv8n (tree, apple, basket).

The network was trained with 120 epochs and an input image resolution of 320x320 pixels. A small dataset (approximately 80 simulated images, 70/30 train/validation split) was used because the simulated scene has low visual

variability. Since images were captured from the same scene, near-duplicate frames may appear in both subsets, so the reported detection performance should not be interpreted as evidence of generalization. A larger dataset with scene-level splitting and an independent test set is planned. Additionally, the detection algorithm was configured to target the nearest object at each stage, for future applications where there might be more than one fruit tree, and to optimize the mobile device's movement path.

IV. MOBILE ROBOTIC SYSTEM MOVEMENT

Fig. 2 shows the mobile robotic system in the Webots simulation environment. It operates in a differential track configuration, which has been found to facilitate movement over other configurations in many situations [17]. Additionally, it has distance sensors located at various points on the robot, identifiable as red laser lines, which allow it to avoid obstacles during movement. The system also has a 3-degree-of-freedom (RRR) manipulator, which, through a camera-distance sensor combination located on the gripper, largely emulates the function of a depth camera, thus allowing it to spatially identify objects of interest using the previously trained model.

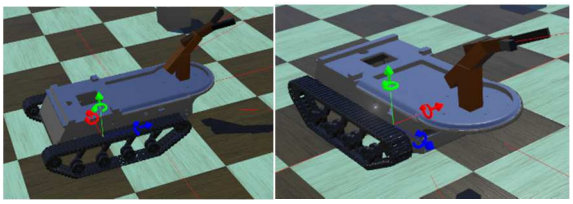


Fig. 2. Tracked mobile platform with 3-DOF manipulator and distance sensors in Webots.

To control the movement of the robotic system, it is modeled as a differential platform with track spacing $L$and linear speed for each track v_L, as well as $v_R$for left and right movement. Equations of motion (1) and (2) allow the conversion from the linear speeds of each wheel to the overall linear $v$and angular speeds $\omega$of the vehicle.

$$v = \frac{v_R + v_L}{2} \tag{1}$$

$$\omega = \frac{v_R - v_L}{L} \tag{2}$$

The mobile platform is in a displacement plane $xz$with respect to the global plane that defines the simulation environment. Here, $y$is the vertical component of height in this coordinate system. Based on this explanation, (3) through (5) allows us to determine the orientation and global position of the mobile platform at any given time, starting from its velocities at a specific moment along with past measurements of orientation and position.

$$\theta_{k+1} = \theta_k + \omega\, dt \tag{3}$$

$$x_{k+1} = x_k + v\sin(\theta_{k+1})\, dt \tag{4}$$

$$z_{k+1} = z_k + v\cos(\theta_{k+1})\, dt \tag{5}$$

Where, in the algorithm used, there is a θ_{k+1} that corresponds to the calculation of the current orientation of the mobile, a θ_k that corresponds to the calculation of the orientation at a past instant, the same example is followed for $x_{k+1}$and z_{k+1}that correspond to the calculation of the current positions of the mobile in the plane, so x_k y z_k This corresponds to the calculation of positions at a past moment. Finally, these calculations also depend on $dt$the sampling time used in Webots; shorter sampling times allow the position of the mobile device to be known at longer points in time but also require greater computing resources.

To move from a current position coordinate pair (x, z) to a desired target coordinate pair, (x_{obj}, z_{obj}) a navigation controller with an attractor vector is used. Simply put, the target coordinates exert an attractive force on the moving object, indicating how and how fast it must move to reach them. Equations (6) to (8) describe the variables of this controller.

$$d_{obj} = \sqrt{(x_{obj} - x)^2 + (z_{obj} - z)^2} \tag{6}$$

$$\varphi_{obj} = \text{atan2}(x_{obj} - x,\, z_{obj} - z) \tag{7}$$

$$e_\theta = (\varphi_{obj} - \theta) \tag{8}$$

Where we have d_{obj} that corresponds to the error in distance between the target position and the current position of the mobile, we also find $e_\theta$that is the orientation error of the mobile. On this basis, an attractive field of proportional control is constructed, which is the one presented in (9) and (10).

$$\nu = K_A\, d_{obj} \tag{9}$$

$$\omega_{atrac} = K_A\, e_\theta \tag{10}$$

Where a linear velocity value is assigned and we obtain an attractive angular velocity component starting from the error in position and orientation multiplied by a gain K_A.

This attractive field controller ensures reaching the target, but it is insufficient when obstacles are present. Therefore, a repulsive field is also used, which operates thanks to distance sensors installed along the moving vehicle. Obstacle avoidance is incorporated as a repulsive angular term ω_{rep} obtained from a weighted combination of the readings from the front and side distance sensors. These are defined as follows: s_1 (front), s_2 (front -left), s_3 (front -right), s_4 (left), s_5 (right). In this way, we generate repulsive pressures as shown in (11).

$$p_i = \text{Umbral} + s_i \tag{11}$$

The lower the sensor reading, the closer it is to the obstacle, and the greater the repulsive force generated by the sensor. Threshold is an iterative adjustment variable. Therefore, the repulsive term is defined as a weighted linear combination of the pressures p_i, as shown in (12).

$$\omega_{\text{rep}} = K_R\,(K_1\, p_1 - K_2\, p_2 + K_3\, p_3 - K_4\, p_4 + K_5\, p_5) \tag{12}$$

This results in a purely angular repulsive component ω_{rep}that only affects the rotation of the object. Finally, the angular velocity imparted to the object is the sum of the attractive angular velocity and the repulsive angular velocity, as shown in (13).

$$\omega = \omega_{\text{atrac}} + \omega_{\text{rep}} \tag{13}$$

In this way, (1) and (2) are applied to transform global speeds into wheel speeds and generate control in the kinematics of the mobile.

V. CONTROL OF THE MANIPULATOR ROBOT FOR TARGET TRACKING AND REACHING

The manipulator, as mentioned, is a three-degree-of-freedom rotational arm (base-shoulder-elbow) plus a gripper, where each link corresponds to a rigid element of the Webots model. The orientation and position of the gripper for grasping the apples are controlled by a visual servoing loop that directly adjusts the joint angles based on the apple's projection error in the image. To achieve this, we obtain the projection errors in the image as shown in (14).

$$e_x = c_x - \frac{W}{2}, e_y = c_y - \frac{H}{2} \tag{14}$$

Where, upon detection of the nearest object of interest, the pixel coordinates of the centroid of said object are extracted $(\mathrm{W}/2, \mathrm{H}/2)$and an error is calculated between said detection and the coordinates of the center of the camera $(c_x\,, c_y)$.

In this way, joint references are updated for a direct proportional joint controller, as shown in (15) to (17).

$$q_{[\text{base}+1]} = q_{\text{base}} - K_{\text{B}}\,\frac{e_x}{W/2} \tag{15}$$

$$q_{[\text{hombro}+1]} = q_{\text{hombro}} + K_{\text{H}}\,\frac{e_y}{H/2} \tag{16}$$

$$q_{[\text{codo}+1]} = q_{\text{codo}} - K_{\text{C}}\,\frac{e_y}{H/2} \tag{17}$$

This allows control over the kinematics of the arm, thus enabling the control of values for the joints. $q_{[\text{base}+1]}$, $q_{[\text{hombro}+1]}$and $q_{[\text{codo}+1]}$, starting from their current valuesq_{base}, $q_{\text{hombro}}$and $q_{\text{codo}}$the errors projected in the image multiplied by gains obtained iteratively, $K_{\text{B}}, K_{\text{H}}\ y\ K_{\text{C}}$ which allow smooth and continuous movements in the links.

The following explains how these target coordinates are obtained (x_{obj}, z_{obj}), which allows the mobile device to be oriented in the correct direction. The overall position of the elements of interest (tree, apples, and basket) is identified by projecting the detection onto the image, along with their absolute distance obtained from the distance sensor, and incorporating the angles reached by the joints, as well as the position and orientation of the mobile device. This process is described below. The pose of the mobile device in its local frame B with respect to the global frame W can be defined as a homogeneous matrix, as shown in (18).

$${}^{W}T_{B}(\mathrm{p}_B) = \begin{bmatrix} {}^{W}R_{B}(\theta_B) & {}^{W}\mathrm{t}_B \\ 0^T & 1 \end{bmatrix} \tag{18}$$

Where there is a vector $\mathrm{p}_B$which is the vector of states associated with the mobile, ${}^{W}R_{B}(\theta_B)$ which is the orientation matrix of the mobile with respect to its global frame and finally ${}^{W}\mathrm{t}_B$which is the translation vector of frame B with respect to W. This pose is obtained from the robot's odometry, which was presented in previous stages. The manipulator is a 3-degree-of-freedom arm, with q-joints as shown in (19).

$$\mathrm{q} = \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} q_{\text{base}} \\ q_{\text{hombro}} \\ q_{\text{codo}} \end{bmatrix} \tag{19}$$

The homogeneous transformation of the frame of effector E with respect to the mobile B corresponds to the combination of transformations with respect to the joints q as shown in (20).

$${}^{B}T_{E}(\mathrm{q}) = {}^{B}T_{1}(q_1)\ {}^{1}T_{2}(q_2)\ {}^{2}T_{3}(q_3) \tag{20}$$

This homogeneous matrix is internally composed of the components shown in (21).

$${}^{B}T_{E}(\mathrm{q}) = \begin{bmatrix} {}^{B}R_{E}(\mathrm{q}) & {}^{B}\mathrm{t}_{E}(\mathrm{q}) \\ 0^T & 1 \end{bmatrix} \tag{21}$$

Where it corresponds ${}^{B}R_{E}(\mathrm{q})$to the orientation of the end effector, i.e., the gripper or claw, with respect to the mobile, and ${}^{B}\mathrm{t}_{E}(\mathrm{q})$is the position of the end effector with respect to the origin of robot frame B. The pose of the effector in the world is obtained by composing with the pose of the mobile, as shown in (22).

$${}^{W}T_{E}(\mathrm{p}_B, \mathrm{q}) = {}^{W}T_{B}(\mathrm{p}_B)\ {}^{B}T_{E}(\mathrm{q}) \tag{22}$$

This composition reflects that the effector moves both due to the movement of the arm, corresponding to variations in the joints, qand to the movement of the mobile, variations of p_B. The camera mounted on the manipulator defines the frame C. Its relationship with the effector is expressed by the transformation matrix shown in (23).

$${}^{E}T_{C} = \begin{bmatrix} {}^{E}R_{C} & {}^{E}\mathrm{t}_{C} \\ 0^T & 1 \end{bmatrix} \tag{23}$$

Where, similarly to matrices shown above, we have an orientation and position component of frame E with respect to frame C. Subsequently, (22) and (23) are integrated to obtain the matrix ${}^{W}T_{C}(\mathrm{p}_B, \mathrm{q})$that allows relating and finding the position and orientation of the camera with respect to the global frame, which depends on the values of the joints and the position of the mobile, as shown in (24).

$${}^{W}T_{C}(\mathrm{p}_B, \mathrm{q}) = {}^{W}T_{E}(\mathrm{p}_B, \mathrm{q})\ {}^{E}T_{C} \tag{24}$$

The elements of interest are detected by the yolov8n network on a 2D image, which delivers the position of the center of the box in pixels, modeled as the coordinate vector u, shown in (25).

$$\mathbf{u} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{25}$$

Where $u$corresponds to the horizontal coordinate in pixels found and $v$corresponds to the vertical coordinate in pixels. In addition, the camera can be described with a matrix of intrinsic parameters, as seen in (26).

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{26}$$

The matrix is composed of $f_x\ y\ f_y$ the effective focal lengths on the camera axes and $c_x\ y\ c_y$ the coordinates of the optical center in the image. To obtain the direction of the ray passing through the camera towards a specific pixel, denormalization is performed as shown in (27).

$$\begin{bmatrix} x_n \\ y_n \\ 1 \end{bmatrix} = K^{-1}\,\mathrm{u} \tag{27}$$

This yields x_n y $y_n$image coordinates in units relative to the image plane; in this case, everything is in meters. Continuing, to identify the coordinates of target detection from the camera's local frame to the global frame, the transformation in (28) is applied.

$$^{W}\mathrm{d}_C = {}^{W}R_C \begin{bmatrix} x_n \\ y_n \\ 1 \end{bmatrix} \tag{28}$$

is obtained here $^{W}\mathrm{d}_C$. This detection, as shown, involves projecting the ordered pair (x_n, y_n) onto the camera's orientation within the global frame. Equation (29) performs a normalization to obtain a unit direction vector.

$$\hat{\mathrm{d}}_C = \frac{^{W}\mathrm{d}_C}{\| \, ^{W}\mathrm{d}_C \, \|} \tag{29}$$

Finally, based on a detection $d$made by the distance sensor on the manipulator that corresponds to the distance measured from the center of the camera to the lens, the global position of the object is estimated with (30).

$$^{W}\mathrm{p}_{\mathrm{obj}} = {}^{W}\mathrm{t}_C + d\,\hat{\mathrm{d}}_C \tag{30}$$

Where $^{W}\mathrm{p}_{\mathrm{obj}} = [x_{obj}, y_{obj}, z_{obj}]^T$ are then the coordinates of the element of interest in the global frame, these coordinates are those used in previous stages for the control of the mobile.

VI. Operating Sequence for Task Execution

As shown in the FSM designed and displayed in Fig. 3, the robot starts from an unknown global point, from which it performs a sweep-type reconnaissance stage in search of its targets. Upon identifying the nearest fruit tree, it obtains its global coordinates and marks it as the search target. It begins navigation using the attraction-repulsion control described earlier until it positions itself at a fixed distance from the platform. Once there, it performs another sweep to find the basket. If it does not find it, it asks the user to provide bin coordinates to proceed to the next stage.

The mobile unit then searches for the nearest apple and marks its coordinates as the target. Once it is close enough for the manipulator's kinematics to allow interaction, it uses the visual loop control described earlier to grasp the apple. If the proximity threshold to the apple cannot be reduced, the detection is marked as unreachable, and the search moves on to the next nearest target. The next step is to move to the container where the apple is deposited. From there, it returns to the tree to find the next apple. The task is marked as complete once all the detected apples on the tree have been deposited in the container or once the remaining apples on the tree are unreachable.

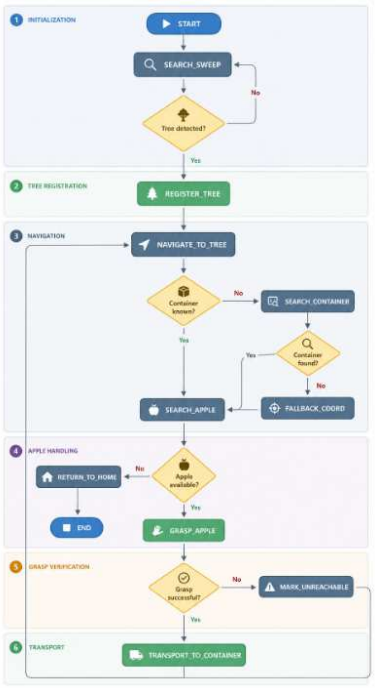


Fig. 3. Finite state machine of the harvesting task.

VII. Results Analysis

Fig. 4 shows the tracking error under the designed visual control, during the detection of an apple where, as can be seen, the values converge to 0 and remain around, during the approach of the mobile to the target.

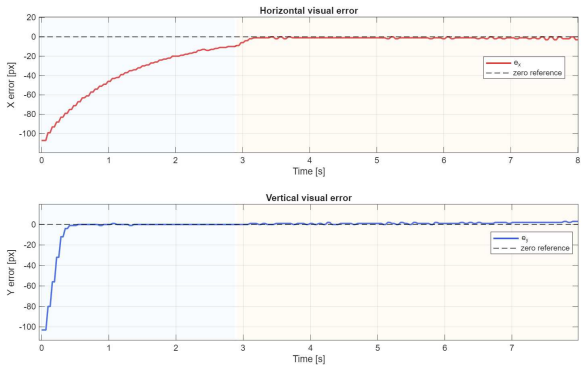


Fig. 4. Image plane tracking errors ex and ey during the approach to an apple.

Since this is a joint movement of the manipulator with the mobile robot, this approach stage generates changes in the commanded angular and linear velocities, as evidenced in Fig. 5. The peak present in the angular velocity is due to the detection occlusion when the apple is very close to the target. The algorithm was configured to ignore values that do not follow a trend with the past commanded values, so there is no negative effect due to this event.

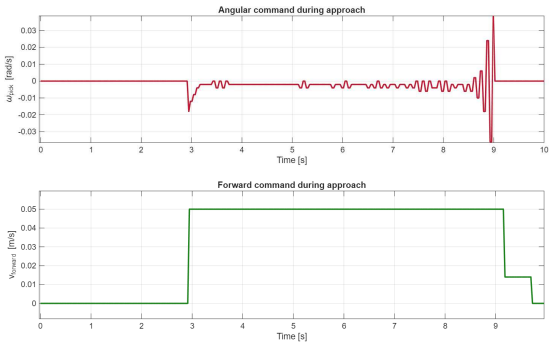


Fig. 5. Commanded linear and angular velocities during the approach to an apple.

Fig. 6 shows the change in joint values due to the designed and implemented kinematic control with visual feedback; note the smoothness with which their values change until the end effector encounters the target.

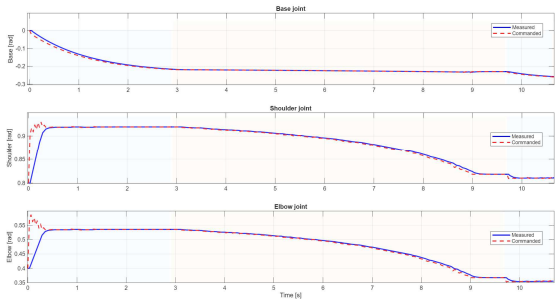


Fig. 6. Evolution of the manipulator joint angles (base, shoulder, elbow) under visual feedback control.

Fig. 7 shows the apple grasping stage, displaying the mobile robot's camera perspective and an external view of the world at the same moment. Note that in this state, the system detects the nearest apple and continuously monitors the manipulator's joints and the robot's movement, thus keeping its center of vision aligned with the target from the moment it is selected until it can grasp it.

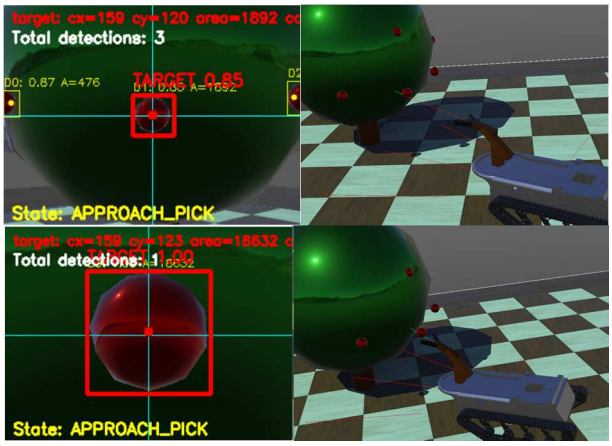


Fig. 7. Grasping stage: robot camera view (left) and external view (right).

The system worked with the routine described above, during which it performed the task continuously and without failure under the conditions used as seen in Fig. 8, where it holds the apples, moves them and deposits them one by one in the place marked for this purpose.

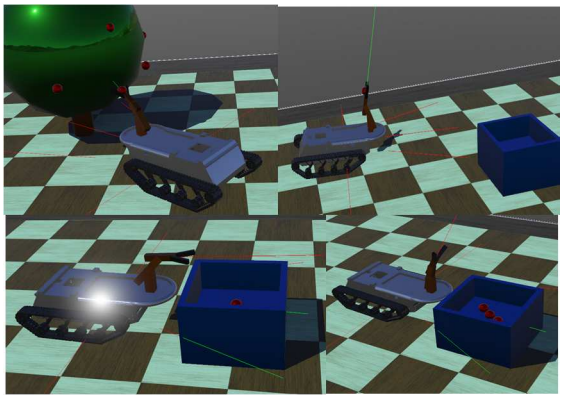


Fig. 8. Transport and deposit of an apple in the basket.

To validate foreseeable exceptions and failures in the system, such as the inability to reach an apple, some apples were placed at heights where the mobile device could see them but not reach them. In these cases, the mobile device approached and positioned the manipulator in the requested configuration; these apples were then marked as unreachable. A proof-of-concept scenario with five apples (three reachable, two intentionally placed out of reach). As shown in Fig. 9, two remained on the tree at the end of the tests because they were not reachable targets. All three reachable apples (3/3) were transferred, the two unreachable apples were correctly flagged and skipped (2/2).

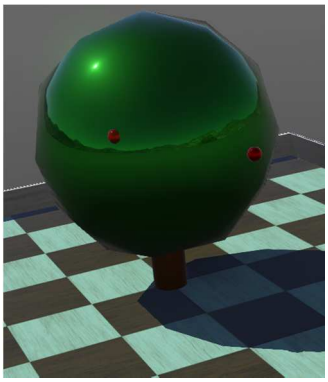


Fig. 9. Final state: three apples transferred to the basket, two unreachable apples remain on the tree.

After conducting multiple tests in the simulation environment, an average distance of 14.7 meters was obtained in an average time of 7.3 minutes. During the execution, delays of approximately 2 minutes of the total time were observed, mainly associated with the approach and placement maneuvers of the elements of interest.

VIII. Conclusions

The results show that the designed system operates without errors in the simulation environment, although it presents an overtime of approximately 14% associated with the security configurations; this margin will be subject to optimization in subsequent developments.

Regarding network performance, average inference times of 25 ms and around 34 FPS on the entire pipeline were obtained during target detection and tracking, confirming the system's ability to perform tasks in real time and operate autonomously, establishing a solid foundation for future validations in real-world environments.

Taken together, the reported metrics support the feasibility of the proposed architecture at a functional level for the detection and transport of items on a mobile platform, showing a solid balance between tracking accuracy, robustness against unknown environments, and operational safety.

Limitations. All results were obtained in Webots, no physical robot, real fruit, outdoor lighting, soil/track slip, contact dynamics, or sensor noise beyond the simulator´s model were evaluated. Consequently, the results demonstrate functional feasibility of architecture, not field robustness.

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